Preliminary

TECHNICAL DATASHEET #TDAX081860

48Vdc/24Vdc Non-isolated Step-Down Converter

720 W Output

P/N: AX081860

- Wide input operating voltage 30 to 72 Vdc (48 Vdc nominal)
- Conditioned output of 24Vdc
- High current output capability (30 A nominal), 720W
- Switch mode operation delivers very high efficiency >96%
- Capable for parallel (redundancy) application
- Design manages heat dissipation avoiding forced cooling
- Reverse polarity, input under/over voltage protections
- Inrush current limit
- Overtemperature protection
- Short-circuit protection
- Rugged and highly reliable
- Compact size for ease of mounting in confined spaces
- Suitable for moist, high shock and vibration environments
- Operational from -40 to 70°C
- IP67 protection



Applications: Designed to interface between 48V batteries and 24V electrical systems found on mobile construction equipment, lift and access equipment and buses.

Ordering Part Numbers:

Converter with Wire Harness KIT: **AX081860K** (Converter, Wire Harness)

Items can also be ordered individually. Converter:
AX0801860
12 pin Mating Wire Harness, 2 m: **AX070163**
Or Mating Plug Kit: **AX070105**

Description: The DC-DC Converter provides regulated 24Vdc power suitable for solenoids, relays and other electrical systems. For operation under the most harsh and demanding conditions, the IP67 rated unit protects against moisture, shock and vibration. Power from a battery or other source of 48 Vdc is converted to a 24Vdc output. The unit has a high current output capability of 30 Amps. The device manages heat dissipation and requires no forced cooling systems. Short-circuit and reverse polarity protections are provided. The compact unit is designed with extremely rugged surge and transient suppression in addition to sustained over/under voltage protection. With a nameplate rating of 720 Watts of output power, the DC-DC Converter provides proprietary multi-phase topology for a high efficiency of >96%.

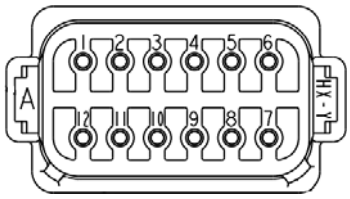
Axiomatic Technologies Corporation
1445 Courtneypark Dr. E.
Mississauga, ON CANADA L5T 2E3
Tel: +1 (905) 602-9270
sales@axiomatic.com
www.axiomatic.com

Axiomatic Technologies Inc.
39100 Country Club Drive, Suite 400
Farmington Hills, MI USA 48331
Tel: +1 (734) 262-8535
sales@axiomatic.com
www.axiomatic.com

Axiomatic Technologies Oy
Höytämöntie 6 C
33880 Lempäälä, FINLAND
Tel: +358 103 375 750
salesfinland@axiomatic.com
www.axiomatic.fi

Technical Specifications:

Input Specifications		**Output Specifications**	
Power Source	48 Vdc nominal	Nameplate Rating (Output Power)	720 Watts nominal
Operating Voltage Range	30 to 72 Vdc	Output Current (DC)	30 A continuous
Maximum Input Current	25 A_{DC} @ 30 VDC, 30 A I-output	Output Voltage	23.5-26.5 Vdc Maximum Factory set at 24 Vdc (nominal) ± 0.5% Regulation ± 3%
Inrush Current Protection	Provided	Output Voltage Ripple	$V_{O(RIPPLE)} \leq$ 100 mVpp
Input Reverse & Under/Over-voltage Protection	Provided Turn-on >23V Turn-off <20V	Start-up Time (with full load)	1 sec. maximum
Output Over-voltage Protection	Provided 80 V Typical	Turn-on Overshoot	<1% of output voltage
		Stability	Stable at all load conditions
Isolation	Not provided		
		Transient Response	5% of Output Voltage (25% - 75% Load, Vin = 48V)
		Short-Circuit Protection	Provided, Auto-recover
		Overload Protection	Provided 35A Typical

General Specifications	
Efficiency	>96% Refer to Figure 1.0.
Quiescent Current	60mA @ 48Vdc Typical
Operating Temperature	-40 to 70°C (-40 to 158°F) Refer to Figure 2.0.
Storage Temperature	-40 to 85°C (-40 to 185°F)
Humidity	0-99% relative humidity (non-condensing)
Shock and Vibration Pending	MIL-STD-202G, Test 204D and 214A (Sine and Random), Test 213BA – 6 ms (Shock) 15 g peak (Sine) 7.68 Grms peak (Random) 50g (Shock)
Enclosure	Cast Aluminum housing, integral gasket and connector 8.14 x 11.12 x 3.11 inches (206.00 x 282.00 x 79.00 mm) L x W x H including integral connectors Refer to the dimensional drawing, Figure 3.0.
Protection rating	IP67
Weight	6.44 lb (2.92 kg)
Electrical Pinout	 12 pin connector (equivalent TE Deutsch P/N: DT13-12P) <u>Connector Pin out:</u> Pin 5: Voltage Input + Pin 6: Voltage Input + Pin 7: Voltage Input + Pin 3: Voltage Input - Pin 4: Voltage Input - Pin 8: Voltage Input - Pin 9: Voltage Output - Pin 10: Voltage Output - Pin 11: Voltage Output - Pin 1: Voltage Output + Pin 2: Voltage Output + Pin 12: Voltage Output + (Input - and Output - are internal connected) Suitable for 14 AWG wire

Input Mating Wire Harness	For the 12-pin input connector, a **mating wire harness, P/N AX070163,** is equivalent to the TE Deutsch P/Ns: DT06-12SA, W12S and twelve contact sockets 0462-201-16141, with 2m (6.5 ft.) of 14 AWG unterminated lead wires. Refer to Figure 4.0. Also available is a plug kit. Mating Plug KIT (DT06-12SA, W12S, 12 0462-201-16141 contacts, 3 sealing plugs) P/N: **AX070105**
Installation	
Set up 1. A 30A fuse is recommended in the primary circuit to provide protection for the primary wiring. 2. Use four ¼-20 1-inch screws to mount the converter. 3. Snap the mating plug connector with wiring harness into the receptacle mounted on the converter. 4. Once the load is ready to receive power, turn on the power source to the converter.	
Grounding	Protective Earth (PE) must be connected to the chassis to reduce the risk of electric shock. All chassis grounding should go to a single ground point designated for the machine and all related equipment.
Mounting	Mounting ledges include holes sized for ¼ inch or M6 bolts. The bolt length will be determined by the end-user's mounting plate thickness. Typically, ¾ inch (20 mm) is adequate. If the module is mounted without an enclosure, it should be mounted vertically with connectors facing left and right to reduce likelihood of moisture entry. All field wiring should be suitable for the operating temperature range of the module. Install the unit with appropriate space available for servicing and for adequate wire harness access (6 inches or 15 cm) and strain relief (12 inches or 30 cm).
Paralleling	The converters can be installed in a parallel configuration for current sharing or redundancy.

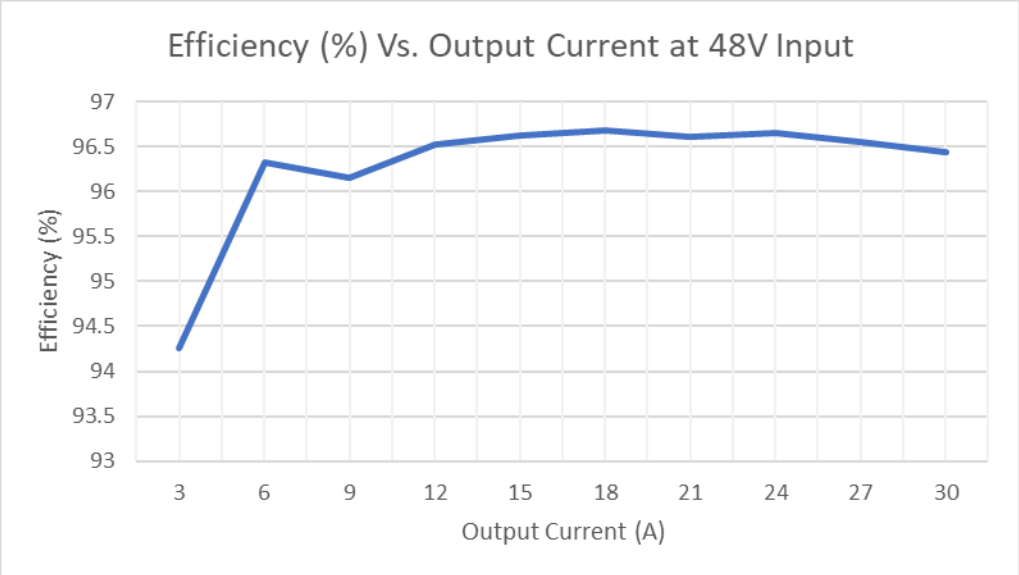


Figure 1.0 - Efficiency at 48V Input and Different Loads

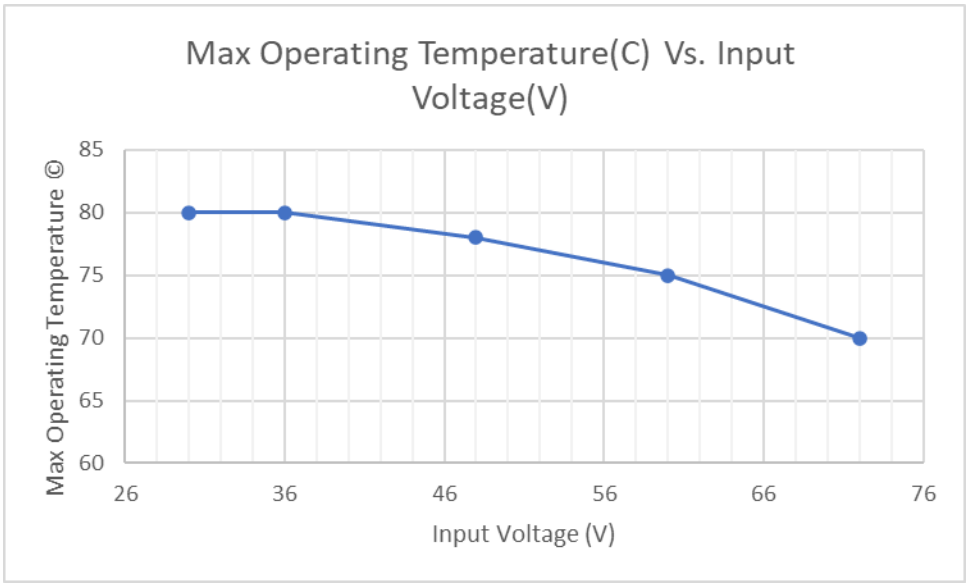


Figure 2.0 - Max. Operating Temperature vs. Input Voltage at 30A Load

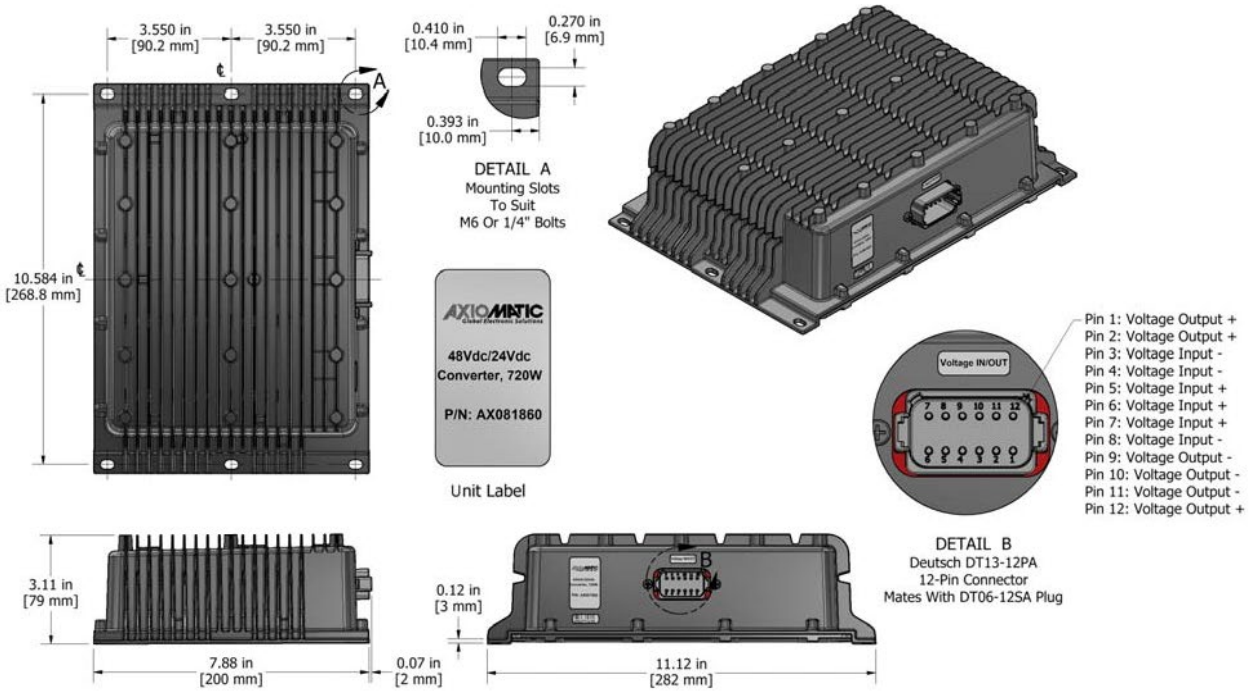


Figure 3. 0. – Dimensional Drawing

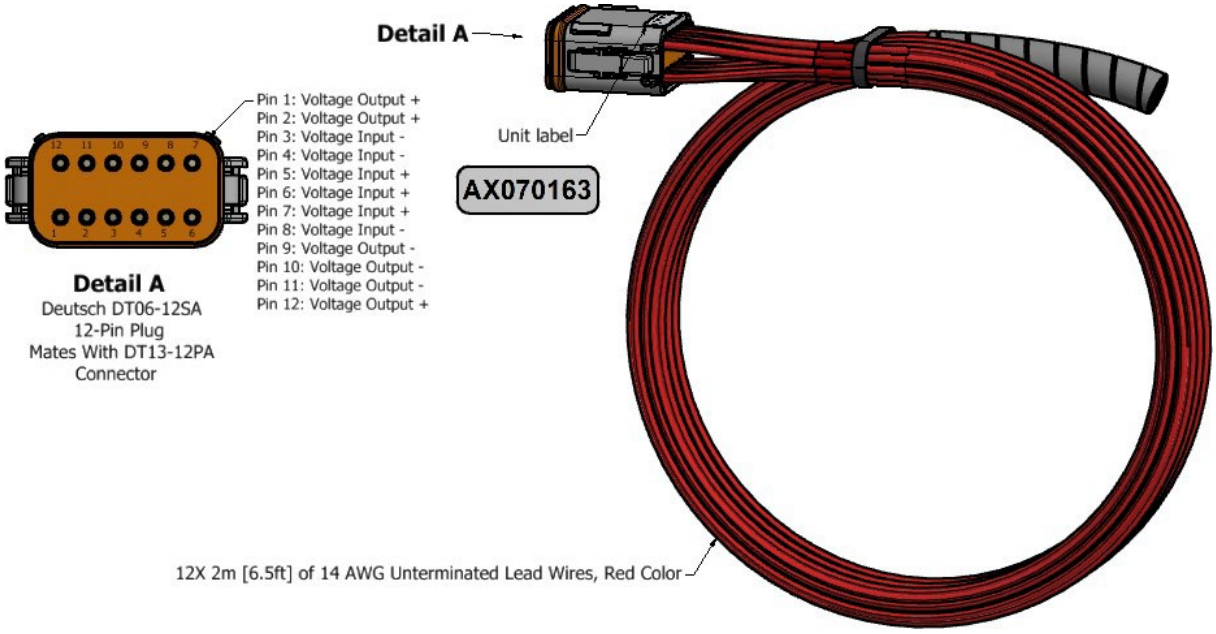


Figure 4.0 – Drawing of Wire Harness AX070163

Specifications are indicative and subject to change. Actual performance will vary depending on the application and operating conditions. Users should satisfy themselves that the product is suitable for use in the intended application.
All our products carry a limited warranty against defects in material and workmanship. Please refer to our Warranty, Application Limitations & Return Materials Process as described on https://www.axiomatic.com/service/.

Form: TDAX081860-08/28/2026

OUR PRODUCTS

AC/DC Power Supplies
Actuator Controls/Interfaces
Automotive Ethernet Switches
Battery Chargers
CAN Controls, Routers, Repeaters
CAN/WiFi
CAN/Bluetooth (Classic, BLE)
CAN bus Routers
CANopen I/O Modules
Current/Voltage/PWM Converters
DC/DC Power Converters
Engine Temperature Scanners
Ethernet/CAN Converters, Gateways
Ethernet Switches
Fan Drive Controllers
Gateways, CAN/Modbus, RS-232
Gyroscopes, Inclinometers
Hydraulic Valve Controllers
Inclinometers, Triaxial
I/O Controls
LVDT Signal Converters
Machine Controls
Media Converters
Modbus, RS-422, RS-485 Controls
Motor Controls, Inverters
Power Supplies, DC/DC, AC/DC
PWM Signal Converters/Isolators
Resolver Signal Conditioners
SAE J1939 I/O Modules
Service Tools
Signal Conditioners, Converters
Strain Gauge CAN Controls
Surge Suppressors

OUR COMPANY

Axiomatic provides electronic machine control components to the off-highway, commercial vehicle, electric vehicle, power generator set, material handling, renewable energy and industrial OEM markets. ***We innovate with engineered and off-the-shelf machine controls that add value for our customers.***

QUALITY DESIGN AND MANUFACTURING

We have an ISO9001 registered R&D, electronics design and manufacturing facility in Canada.

Sales offices are in Canada, USA, Australia and Finland.

COMPLIANCE

Product compliance details can be found in the product literature and/or on axiomatic.com. Any inquiries should be sent to sales@axiomatic.com.

SAFE USE

All products should be serviced by Axiomatic. Do not open the product and perform the service yourself.



This product can expose you to chemicals which are known in the State of California, USA to cause cancer and reproductive harm. For more information go to www.P65Warnings.ca.gov.

TECHNICAL SUPPORT

Contact technical support by emailing support@axiomatic.com.

WARRANTY, APPLICATION APPROVALS/LIMITATIONS

Axiomatic Technologies Corporation reserves the right to make corrections, modifications, enhancements, improvements, and other changes to its products and services at any time and to discontinue any product or service without notice. Customers should obtain the latest relevant information before placing orders and should verify that such information is current and complete. Users should satisfy themselves that the product is suitable for use in the intended application. All our products carry a limited warranty against defects in material and workmanship. Please refer to our Warranty, Application Approvals/Limitations and Return Materials Process at https://www.axiomatic.com/service/.

SERVICE

All products to be returned to Axiomatic require a Return Materials Authorization Number (RMA#) from rma@axiomatic.com. Please provide the following information when requesting an RMA number:

- Serial number, part number, Runtime hours, description of problem
- Wiring set up diagram, application and other comments as needed

DISPOSAL

Axiomatic products are electronic waste. Please follow your local environmental waste and recycling laws, regulations and policies for safe disposal or recycling of electronic waste.

CONTACTS

Axiomatic Technologies Corporation
1445 Courtneypark Drive E.
Mississauga, ON
CANADA L5T 2E3
TEL: +1 (905) 602-9270
FAX: +1 (905) 602-9279
www.axiomatic.com
sales@axiomatic.com
support@axiomatic.com

Axiomatic Technologies Oy
Höytämöntie 6 C
33880 Lempäälä
FINLAND
TEL: +358 10 337 5751
www.axiomatic.com
salesfinland@axiomatic.com

Axiomatic Technologies Corporation
Unit 1632
3 Lakeview Rise
Noosa Heads, QLD
AUSTRALIA 4567
TEL: +61 (0) 423 600 664
www.axiomatic.com
sales@axiomatic.com

Axiomatic Technologies Inc.
39100 Country Club Drive
Suite 400
Farmington Hills, MI
USA 48331
TEL: +1 (734) 262-8535
www.axiomatic.com
sales@axiomatic.com

Copyright 2026